C4: closed-form time integrals of the second-order radiation-era solutions (Poisson gauge, $a \propto \eta$)

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1 Setup and result in one paragraph

Conventions of SPEC.md and report_C1: $X = \epsilon X_1 + \epsilon^2 X_2/2$, $X_{2,\mathbf{k}} = \int \frac{d^3q}{(2\pi)^3} K_X(q, |\mathbf{k}-\mathbf{q}|; \eta)\, \zeta_{\mathbf{q}} \zeta_{\mathbf{k}-\mathbf{q}}$, $p = |\mathbf{k}-\mathbf{q}|$, $x_a = a\eta/\sqrt{3}$, $T_\Phi(x) = -2(\sin x - x\cos x)/x^3$. The only non-algebraic second-order time integrals are

$$\Phi_2 = P(\eta) + K_{\Phi_2,0}\, \hat{g}(k\eta), \qquad P = \int_0^\eta G_\Phi(\eta,\eta') S_\Phi(\eta') d\eta', \qquad \hat{g} = 3u_1 = -\tfrac{3}{2} T_\Phi(x_k), \tag{1}$$

$$h_{2,ij} = \Lambda_{ij,kl} q_k q_l\, I_h, \qquad I_h = \int_0^\eta \frac{\eta'}{\eta} \frac{\sin k(\eta-\eta')}{k} K_h(\eta') d\eta' = \frac{4}{k^2} I_{\rm RD}^{\rm KT}\left(\tfrac{p}{k}, \tfrac{q}{k}, k\eta\right). \tag{2}$$

Both are obtained in CLOSED FORM: elementary functions plus $\mathrm{Si}(w\eta)$ and the entire function $\mathrm{Cin}(z) = \int_0^z (1-\cos t)\, dt/t = \gamma_E + \ln|z| - \mathrm{Ci}(|z|)$ at the frequencies

$$\Phi_2:\ w_1 = \tfrac{k+q+p}{\sqrt{3}},\ w_2 = \tfrac{k+q-p}{\sqrt{3}},\ w_3 = \tfrac{k-q+p}{\sqrt{3}},\ w_4 = \tfrac{q+p-k}{\sqrt{3}}\ (\text{all}\ \geq 0); \qquad h_2:\ k \pm \tfrac{q\pm p}{\sqrt{3}}.$$

All other second-order fields are algebraic in Φ_2, Φ_2' and first-order transfer functions (Sec. **??**); hence **nothing remains as an unevaluated integral at second order**. Because Cin is entire and even, the forms need no $|\cdot|$, principal values or branch choices; rewriting Cin $\to$ Ci reproduces Kohri–Terada (KT) eq. (22) with its $\log|\ldots|$ terms.

2 Method

The integrands are finite sums $\sum c_{n,w}(k,q,p)\, \eta'^n e^{iw\eta'}$, $-7 \leq n \leq 0$. Each term is integrated by $I_n = [t^{n+1}e^{iwt}]/(n+1) - \frac{iw}{n+1} I_{n+1}$, $I_{-1} = -\mathrm{Cin}(w\eta) + i\,\mathrm{Si}(w\eta) + \ln(\eta/\varepsilon)$, on $[\varepsilon, \eta]$. All ε^{-m} and $\ln\varepsilon$ pieces cancel identically (symbolic check) — this is the regularity of the total integrand at $\eta' = 0$. The source of Φ_2 is

$$S_\Phi = K_{S\Phi} = \left[\frac{-7k^6 - 19k^4p^2 - 19k^4q^2 + 7k^2p^4 - 22k^2p^2q^2 + 7k^2q^4 + 3p^6 - 3p^4q^2 - 3p^2q^4 + 3q^6}{12k^4}\right] T_p T_q \tag{3}$$

$$+ \left[\frac{\eta\left(k^2+q^2\right)\left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{12k^4}\right] T_q \dot{T}_p \tag{4}$$

$$+ \left[\frac{\eta\left(k^2+p^2\right)\left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{12k^4}\right] T_p \dot{T}_q \tag{5}$$

$$+ \left[\frac{-\eta^2k^6 - 2\eta^2k^4p^2 - 2\eta^2k^4q^2 + 3\eta^2k^2p^4 - 6\eta^2k^2p^2q^2 + 3\eta^2k^2q^4 + 12k^4 - 24k^2p^2 - 24k^2q^2 + 36p^4 - 72p^2q^2 + 36q^4}{12k^4}\right] \dot{T}_p \dot{T}_q \tag{6}$$

with $T_a \equiv T_\Phi(x_a)$, $\dot{T}_a \equiv \partial_\eta T_\Phi(x_a)$, and $G_\Phi = [u_1(\eta')u_2(\eta) - u_1(\eta)u_2(\eta')]/W(\eta')$, $u_1 = (\sin x - x\cos x)/x^3$, $u_2 = (\cos x + x\sin x)/x^3$, $W = -3\sqrt{3}/(\eta^4 k^3)$.

3 Structure of Φ_2

$$P = u_2(x_k)\,\mathcal{A}(\eta) - u_1(x_k)\,\mathcal{B}(\eta), \qquad \mathcal{A} = \int_0^\eta \frac{u_1}{W} S_\Phi, \quad \mathcal{B} = \int_0^\eta \frac{u_2}{W} S_\Phi. \tag{7}$$

The special-function skeleton is governed by ONE coefficient,

$$\mathcal{D}(k,q,p) = \frac{(k^2+p^2+q^2)\,R}{8k^2p^3q^3}, \quad R = 3k^6 - 3k^4(p^2+q^2) - 3k^2(p^4+q^4) - 2k^2p^2q^2 + 3(p^6+q^6) - 3p^2q^2(p^2+q^2), \tag{8}$$

$$\mathcal{A} \supset \mathcal{D}\,[\mathrm{Si}(w_1\eta) - \mathrm{Si}(w_2\eta) - \mathrm{Si}(w_3\eta) - \mathrm{Si}(w_4\eta)], \qquad \mathcal{B} \supset -\tfrac{\mathcal{D}}{1}\,[\ldots] \text{ (see Appendix)} \tag{9}$$

the rest of $\mathcal{A}, \mathcal{B}$ being trigonometric polynomials times Laurent polynomials in η (full form: Appendix **??**). Only the $n = 0$ monomials produce $1/w$ factors; they become removable $0/0$ at the collinear edges $w_2 = 0$ ($\mu = -1$), $w_3 = 0$ ($\mu = 1, q > k$), $w_4 = 0$ ($\mu = 1, q < k$). Exact collinear limiting forms are obtained by substituting $p = k + q,\ q - k,\ k - q$ in the integrand before integrating (`collinear_closed.pkl`); zero-frequency terms occur only at $n = -4, -2, 0$, so no $\ln\eta$ appears in collinear Φ_2. No $\sqrt{3}$-resonance exists for Φ_2 (sound speed enters both G_Φ and the source).

4 Tensor control: h_2

$I_h = [\sin(k\eta)A_h(\eta) - \cos(k\eta)B_h(\eta)]/\eta$ with $A_h, B_h = \mathrm{Cin/Si}$ at $k \pm (q \pm p)/\sqrt{3}$ plus elementary terms (Appendix **??**). It equals $(4/k^2)\times\mathrm{KT}(22)$ [with $|\cdot|$ inside its log] to 10^{-15} (double) / 25 digits (exact inputs). Resonance $q+p = \sqrt{3}k$: harmless in Cin form; the exact resonant form (separately integrated) contains an explicit $\ln\eta$. The other resonance $|q-p| = \sqrt{3}k$ lies outside the physical domain.

5 All other fields (algebraic)

$$\Psi_2 = \Phi_2 - Q_2, \qquad Q_2 = K_Q = \left[-\frac{13k^4 - 6k^2p^2 - 6k^2q^2 + 9p^4 - 18p^2q^2 + 9q^4}{4k^4}\right] T_pT_q \tag{10}$$

$$+ \left[-\frac{\eta\left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{4k^4}\right] T_q\dot{T}_p \tag{11}$$

$$+ \left[-\frac{\eta\left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{4k^4}\right] T_p\dot{T}_q \tag{12}$$

$$+ \left[-\frac{\eta^2\left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{4k^4}\right] \dot{T}_p\dot{T}_q \tag{13}$$

$$\delta_2 = \tfrac{\eta^2}{3}K_{S00} - \tfrac{2}{3}\eta^2k^2\Phi_2 - 2\eta\Phi_2' - 2\Psi_2, \qquad \delta_{T2} = \tfrac{1}{4}\delta_2 - \tfrac{3}{16}T_\delta(q)T_\delta(p), \tag{14}$$

$$V_2 = -\tfrac{\eta^2}{4}\left[2\big(\Phi_2' + \Psi_2/\eta\big) + \frac{K_{S0z}}{ik}\right], \quad v_{2,i} = \partial_i V_2 + v_{2,i}^T, \tag{15}$$

$$\omega_{2,i} = P_{ij}(\hat{k})q_j\,K_\omega, \qquad K_\omega = \left[-\frac{2i\eta\,(p-q)\,(p+q)}{k^2}\right] T_pT_q \tag{16}$$

$$+ \left[\frac{2i\eta^2q^2}{k^2}\right] T_q\dot{T}_p \tag{17}$$

$$+ \left[-\frac{2i\eta^2p^2}{k^2}\right] T_p\dot{T}_q, \tag{18}$$

$$v_{2,i}^T = P_{ij}(\hat{k})q_j\,K_{vT} \quad \text{(explicit in second_order.pkl; } K_{vT} = -(\eta^2k^2/8 + 1)K_\omega - \tfrac{\eta^2}{4}K_{S0\perp}). \tag{19}$$

K_{S00}, K_{S0z} are C1's (closed in T_Φ); Φ_2' is the closed-form derivative ($G_\Phi(\eta,\eta) = 0$).

6 Limits

$\eta \to 0$ **(Frobenius + constraints; checked against the closed forms at 120 digits)**

$$\Phi_2 = \left[-\frac{7k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4}{9k^4}\right]\eta^0 \tag{20}$$

$$+ \left[\frac{-21k^4p^2 - 21k^4q^2 + 10k^2p^4 - 28k^2p^2q^2 + 10k^2q^4 + 3p^6 - 3p^4q^2 - 3p^2q^4 + 3q^6}{270k^4}\right]\eta^2 \tag{21}$$

$$+ \left[-\frac{-150k^6p^2 - 150k^6q^2 - 65k^4p^4 - 398k^4p^2q^2 - 65k^4q^4 + 80k^2p^6 - 144k^2p^4q^2 - 144k^2p^2q^4 + 80k^2q^6 + 15p^8 + 36p^6q^2 -}{113400k^4}\right] \tag{22}$$

$$I_h = \left[\frac{8}{9}\right]\eta^2 \tag{23}$$

$$+ \left[-\frac{2\left(3k^2 + p^2 + q^2\right)}{135}\right]\eta^4 + O(\eta^6) \tag{24}$$

$$\Psi_2 = \left[\frac{2\left(3k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{9k^4}\right]\eta^0 \tag{25}$$

$$+ \left[-\frac{2\left(8k^4p^2 + 8k^4q^2 - 5k^2p^4 + 2k^2p^2q^2 - 5k^2q^4 + 3p^6 - 3p^4q^2 - 3p^2q^4 + 3q^6\right)}{135k^4}\right]\eta^2 \tag{26}$$

$$+ \left[\frac{75k^6p^2 + 75k^6q^2 + 55k^4p^4 + 234k^4p^2q^2 + 55k^4q^4 - 75k^2p^6 - 117k^2p^4q^2 - 117k^2p^2q^4 - 75k^2q^6 + 45p^8 + 108p^6q^2 - 306p}{56700k^4}\right] \tag{27}$$

$$\delta_2 = \left[-\frac{4\left(-5k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{9k^4}\right]\eta^0 \tag{28}$$

$$+ \left[\frac{2\left(15k^6 - 41k^4p^2 - 41k^4q^2 - 5k^2p^4 + 2k^2p^2q^2 - 5k^2q^4 + 3p^6 - 3p^4q^2 - 3p^2q^4 + 3q^6\right)}{135k^4}\right]\eta^2 \tag{29}$$

$$+ \left[-\frac{-1095k^6p^2 - 1095k^6q^2 - 275k^4p^4 - 3394k^4p^2q^2 - 275k^4q^4 - 25k^2p^6 - 39k^2p^4q^2 - 39k^2p^2q^4 - 25k^2q^6 + 15p^8 + 36p^6q^2}{28350k^4}\right] \tag{30}$$

$$V_2 = \left[-\frac{7k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4}{9k^4}\right]\eta^1 \tag{31}$$

$$+ \left[\frac{39k^4p^2 + 39k^4q^2 + 6k^2p^4 - 20k^2p^2q^2 + 6k^2q^4 + 3p^6 - 3p^4q^2 - 3p^2q^4 + 3q^6}{270k^4}\right]\eta^3 \tag{32}$$

$$+ \left[-\frac{375k^6p^2 + 375k^6q^2 + 105k^4p^4 + 134k^4p^2q^2 + 105k^4q^4 + 105k^2p^6 - 169k^2p^4q^2 - 169k^2p^2q^4 + 105k^2q^6 + 15p^8 + 36p^6q^2}{113400k^4}\right] \tag{33}$$

$\eta \to \infty$

With $\Lambda = \ln \frac{(k+q+p)(q+p-k)}{(k+q-p)(k+p-q)}$:

$$\Phi_2 \to \frac{1}{\eta^2}\Big\{C_L\left[\Lambda\cos x_k + \pi \sin x_k\right] + C_0 \cos x_k\Big\} + O(\eta^{-3}), \qquad C_L = -\frac{3(k^2+p^2+q^2)R}{8k^4p^3q^3}, \tag{34}$$

$$C_0 = \frac{3\left[-3k^6 - 3k^4(p^2+q^2) + 3k^2(p^4+q^4) + 2k^2p^2q^2 + 3(p^6+q^6) - 3p^2q^2(p^2+q^2)\right]}{2k^4p^2q^2}, \tag{35}$$

$$I_h \to \frac{4}{k^2}\,\mathrm{KT}(25) \quad (\sin k\eta \text{ log term} + \Theta(q+p-\sqrt{3}k)\,\pi\cos k\eta \text{ term}). \tag{36}$$

Next orders (η^{-3} for Φ_2, η^{-2} for I_h) in `asymptotics.pkl`; verified up to $k\eta = 10^5$ (remainder scales as the first dropped power). The expansion is non-uniform as $q \to 0$ or $p \to 0$ and near the collinear edges / $q + p = \sqrt{3}k$.

7 Verification

- Quadrature arbiter (Gauss–Legendre, 50–60 digits) vs closed forms: P at 11 points incl. collinear and $\sqrt{3}$ points: $\leq 3.5 \times 10^{-40}$; I_h: 25 digits; Φ_2, Φ_2', I_h at 10 random points in `verify_module.py`.
- END-TO-END: all C4 fields inserted into C1's full second-order Einstein tensor (10 components) and the 4 conservation equations for a plane-wave pair: residual/max term $\leq 5.5\times10^{-50}$ at 3 points ($k\eta = 2.5$, 9.6, 20.1).
- ODE residuals by substitution; small-η series = Frobenius series; KT (22),(25) controls.
- Fast double-precision evaluators (`second_order_closed.py`) vs 50-digit reference: see report.

A Full form of P (particular integral of Φ_2)

(not generated)

B Full form of I_h

Special-function part (keys: function, frequency index j in the list above):

$$[Cin,\ w_2]:\ -\frac{3\left(-3k^2+p^2+q^2\right)^2\sin(\eta k)}{\eta k p^3 q^3}$$

$$[Cin,\ w_3]:\ -\frac{3\left(-3k^2+p^2+q^2\right)^2\sin(\eta k)}{\eta k p^3 q^3}$$

$$[Cin,\ w_1]:\ \frac{3\left(-3k^2+p^2+q^2\right)^2\sin(\eta k)}{\eta k p^3 q^3}$$

$$[Cin,\ w_4]:\ \frac{3\left(-3k^2+p^2+q^2\right)^2\sin(\eta k)}{\eta k p^3 q^3}$$

$$[Si,\ w_1]:\ \frac{3\left(-3k^2+p^2+q^2\right)^2\cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)}{\eta k p^3 q^3 \mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)}$$

$$[Si,\ w_4]:\ \frac{3\left(\mathrm{re}\left(\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)+\mathrm{re}\left(\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)(-3k$$
$$2\eta k p^3 q^3 \mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)$$

$$[Si,\ w_3]:\ -\frac{3\left(\mathrm{re}\left(\mathrm{Si}\left(-\eta k-\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)+\mathrm{re}\left(\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k-\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)(-3$$
$$2\eta k p^3 q^3 \mathrm{Si}\left(-\eta k-\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)$$

$$[Si,\ w_2]:\ -\frac{3\left(\mathrm{re}\left(\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)+\mathrm{re}\left(\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)(-3$$
$$2\eta k p^3 q^3 \mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)$$

Elementary part ($x_a = a\eta/\sqrt{3}$, and $\sin k\eta, \cos k\eta$):

$$\begin{aligned}
&\cos^2(k\eta)\cos(x_p)\cos(x_q)\left\{\left[\frac{72}{p^2q^2}\right]\eta^{-2}\right\}\\
&+\sin(k\eta)\left\{\left[-\frac{12\left(-3k^2+p^2+q^2\right)}{kp^2q^2}\right]\eta^{-1}\right\}\\
&+\sin^2(k\eta)\cos(x_p)\cos(x_q)\left\{\left[\frac{72}{p^2q^2}\right]\eta^{-2}\right\}\\
&+\sin^2(k\eta)\sin(x_p)\cos(x_q)\left\{\left[-\frac{72\sqrt{3}}{p^3q^2}\right]\eta^{-3}\right\}
\end{aligned}$$

$$
\begin{aligned}
&+\sin^2(k\eta)\sin(x_p)\sin(x_q)\Bigg\{\left[\frac{216}{p^3q^3}\right]\eta^{-4}\\
&\qquad+\left[\frac{36\left(-3k^2+p^2+q^2\right)}{p^3q^3}\right]\eta^{-2}\Bigg\}\\
&+\sin^2(k\eta)\sin(x_q)\cos(x_p)\Bigg\{\left[-\frac{72\sqrt{3}}{p^2q^3}\right]\eta^{-3}\Bigg\}\\
&+\sin(x_p)\cos^2(k\eta)\cos(x_q)\Bigg\{\left[-\frac{72\sqrt{3}}{p^3q^2}\right]\eta^{-3}\Bigg\}\\
&+\sin(x_p)\sin(x_q)\cos^2(k\eta)\Bigg\{\left[\frac{216}{p^3q^3}\right]\eta^{-4}\\
&\qquad+\left[\frac{36\left(-3k^2+p^2+q^2\right)}{p^3q^3}\right]\eta^{-2}\Bigg\}\\
&+\sin(x_q)\cos^2(k\eta)\cos(x_p)\Bigg\{\left[-\frac{72\sqrt{3}}{p^2q^3}\right]\eta^{-3}\Bigg\}
\end{aligned}
$$